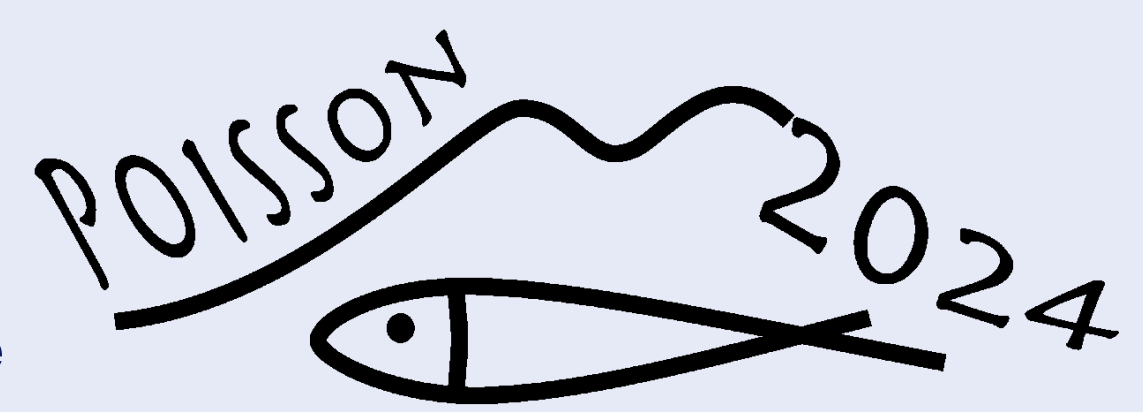




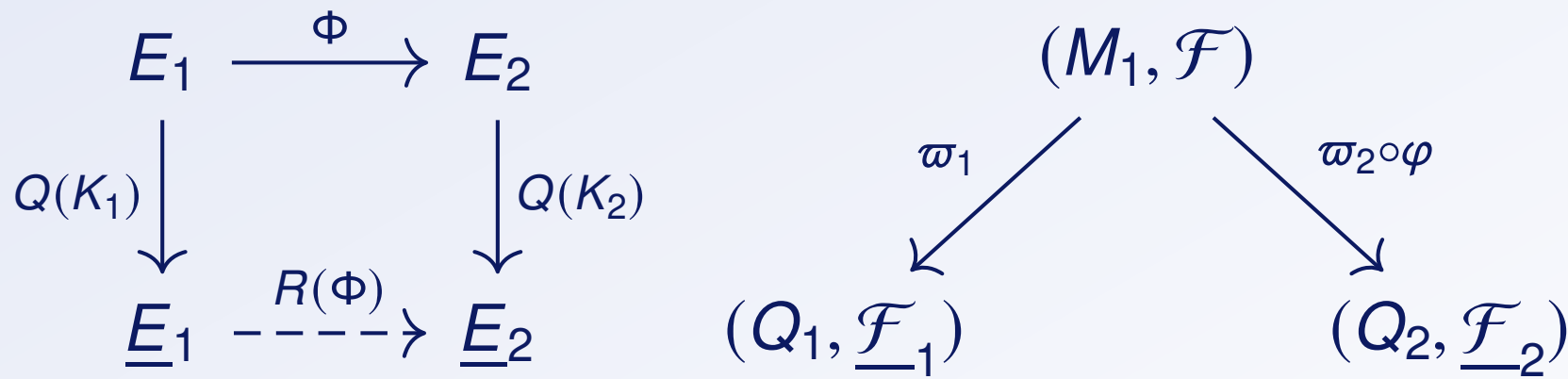
Courant Algebroid Relations

Definition [Vys20]
Let $E_1 \rightarrow M_1$ and $E_2 \rightarrow M_2$ be Courant algebroids (CAs). A **Courant Algebroid Relation (CAR)** $R: E_1 \dashrightarrow E_2$ is an isotropic, involutive subbundle $R \subset E_1 \times \overline{E_2}$ supported on a submanifold $C \subset M_1 \times M_2$, such that $\rho(R^\perp) \subseteq TC$.

- Examples:**
- **Graph of Courant Automorphisms:** if $\Phi: E_1 \rightarrow E_2$ covers $\varphi: M_1 \rightarrow M_2$, then $\text{gr}(\Phi) \subset E_1 \times \overline{E_2}$ is a CAR supported on $\text{gr}(\varphi)$.
 - **Reduction of Courant algebroids à la Zambon:** if $K \subset E$ is isotropic, then $\underline{E} := K^\perp / \rho(K)$ is a CA over $Q = M / \rho(K)$. If $\mathfrak{h}: K^\perp \rightarrow \underline{E}$ is the quotient, then $Q(K) := \text{gr}(\mathfrak{h})$ is a CAR supported on $\text{gr}(\varpi)$, where $\varpi: M \rightarrow Q$ is the quotient map.

T-Duality

A CAR is a **T-duality relation**, denoted $R(\Phi)$, if there are the (not necessarily commutative) diagrams



where $T\mathcal{F} = \rho_1(K_1) + \varphi^* \rho_2(K_2)$ and $[\Gamma(\rho_1(K_1)), \Gamma(\varphi^* \rho_2(K_2))] \subset \Gamma(\rho_1(K_1)) + \Gamma(\varphi^* \rho_2(K_2))$.

Theorem – Existence of T-Duality Relation [DMS23]
 $R(\Phi) = Q(K_2) \circ \text{gr}(\Phi) \circ Q(K_1)^T$ exists whenever $\Phi(K_1) \cap K_2$ has constant rank, and its support $C = \text{gr}(\varpi_2 \circ \varphi) \circ \text{gr}(\varpi_1)^T$ is smooth. $R(\Phi)$ is a Dirac structure.

Remark: C is always smooth if Q_1 and Q_2 are fibred over the same manifold $\mathcal{B} = Q_1 / \underline{\mathcal{F}}_1 = Q_2 / \underline{\mathcal{F}}_2$.

Generalised Isometries

Definition [Vys20]
A CAR $R: (E_1, \tau_1) \dashrightarrow (E_2, \tau_2)$ is a **generalised isometry** if $(\tau_1 \times \tau_2)(R) = R$, where τ_1, τ_2 are generalised metrics.

Theorem – Exist. and Uniq. of T-Dual Gen. Metrics [DMS23]
For exact CAs E_1, E_2 , given an “invariant w.r.t. $\underline{\mathcal{F}}_1$ ” generalised metric τ_1 on $\underline{E}_1$, there exists a unique generalised metric τ_2 on $\underline{E}_2$ such that $R(\Phi)$ is a generalised isometry iff $\Phi(K_1) \cap K_2^\perp \subset K_2$.

Example: Take Q_1, Q_2 to be S^1 -bundles over a manifold $\mathcal{B}$, and let $M = Q_1 \times_{\mathcal{B}} Q_2$. Take connections θ_i on Q_i and dual vector fields ∂_{θ_i} and $\Phi = e^{\theta_1 \wedge \theta_2}$. Then, pointwise,
$$R(e^{\theta_1 \wedge \theta_2}) = \text{Span}\{(X, X), (\xi, \xi), (\theta_1, \partial_{\theta_2}), (\partial_{\theta_1}, \theta_2): X + \xi \in T\mathcal{B}\}$$

supported on M . If the S^1 -invariant τ_1 is given by (g_1, b_1) , then τ_2 given by (g_2, b_2) satisfies the Buscher rules, as in [CG10].

Divergences and Ricci Flow

A **divergence** on a CA E is an operator $\text{div}: \Gamma(E) \rightarrow C^\infty(M)$ satisfying $\text{div}(fe) = f \text{div} e + \mathcal{L}_{\rho(e)} f$, for all $e \in \Gamma(E)$ and $f \in C^\infty(M)$.

Definition
Let $R: E_1 \dashrightarrow E_2$ be a CAR, divergences div_1 and div_2 are **R-related** ($\text{div}_1 \sim_R \text{div}_2$) if $\text{div}_1 e_1 = \text{div}_2 e_2$ for any $(e_1, e_2) \in \Gamma(E_1 \times E_2; R)$.

Theorem – Exist. and Uniq. of T-dual Divergences [DMS24]
If $R(\Phi): (\underline{E}_1, \tau_1) \dashrightarrow (\underline{E}_2, \tau_2)$ is a generalised isometry, div_1 is a divergence on $\underline{E}_1$ invariant wrt $\mathcal{F}_1$, and $\Phi^{-1}(K_2) \cap K_1 = \{0\}$, there is a unique diveregence div_2 on $\underline{E}_2$ such that $\text{div}_1 \sim_{R(\Phi)} \text{div}_2$.

A **dilaton** on a CA E with metric τ is a divergence div_ψ such that
$$\text{div}_\psi e - \mu_\tau^{-1} \mathcal{L}_{\rho(e)} \mu_\tau = \langle \rho^*(d\psi), e \rangle$$

Dilaton shift: Let $R(\Phi): (E_1, \tau_1) \dashrightarrow (E_2, \tau_2)$ be a T-duality relation, div_{ψ_1} a dilaton on E_1 , and $\text{div}_{\psi_1} \sim_{R(\Phi)} \text{div}_2$, then $\langle e_2, \cdot \rangle = \text{div}_2 - \text{div}_{\tau_2}$ is the **dilaton shift**. In the S^1 -bundle case, $e_2 = d(\psi_1 - \frac{1}{2} \log g_{\theta_1 \theta_1})$.

Theorem – Related Ricci Tensors [DMS24]
Let $R: (E_1, \tau_1) \dashrightarrow (E_2, \tau_2)$ be a generalised isometry and $\text{div}_1 \sim_R \text{div}_2$, then $\text{GRic}_{(\tau_1, \text{div}_1)} \sim_R \text{GRic}_{(\tau_2, \text{div}_2)}$, where (as in [ŠV20])
$$\text{GRic}_{(\tau, \text{div})} := \text{div}[v_+, w_-]_+ - w_- \cdot \text{div} v_+ - \text{Tr}_{V^+}([\cdot, w_-]_-, v_+)_+.$$

Corollary – Ricci Flow [DMS24]
If $\tau_1(t)$ is a solution of the generalised Ricci flow, then so is $\tau_2(t)$.

Spinor Relations and Twisted Cohomology

For a CAR $R: E_1 \dashrightarrow E_2$, the Clifford algebra $Cl(R)$ is well defined since $R \subset E_1 \times \overline{E_2}$ is isotropic: $(e_1, e_2) \cdot (e_1, e_2) = \langle e_1, e_1 \rangle = \langle e_2, e_2 \rangle$. There are two actions of $Cl(R)$ on $\wedge^\bullet T^* M_1 \times \wedge^\bullet T^* M_2$:

$$(e_1, e_2) \triangleright^\pm (\rho_1, \rho_2) = (e_1 \triangleright \rho_1, \pm e_2 \triangleright \rho_2)$$

Definition
A subbundle $U_R \subset \wedge^\bullet T^* M_1 \times \wedge^\bullet T^* M_2$ supported on C is a **R-Clifford relation of parity $\pm$** if it is a fibrewise spinor representation of $Cl(R)$ for the action $\triangleright^\pm$.

Theorem – Uniq. of T-Duality Spinor Relations [DMS]
If the leaves of $(Q_2, \underline{\mathcal{F}}_2)$ are orientable, there is a unique (up to scaling) $R(\Phi)$ -Clifford relation $U_{R(\Phi)}$ of parity $(-1)^{\dim \mathcal{F}_2}$ for a T-duality relation $R(\Phi)$.

Explicitly, $U_{R(\Phi)}$ is given by

$$U_{R(\Phi)} = \frac{Cl(R)}{Cl(R)L} \cong \wedge^\bullet F \triangleright^\pm (1, \lambda)$$

for $\lambda \in \det(T^* \mathcal{F}_2)$ a pure spinor in U_R , L the corresponding Lagrangian, and F dual to L such that $\rho_1(p_1(F)) = \{0\}$.

Example: In the case of S^1 -bundles

$$U_{R(\Phi)} = \text{gr} \left(\rho \mapsto \frac{1}{2\pi} \int_{S^1} e^{\theta_1 \wedge \theta_2} \wedge \rho, \quad \rho \in \Omega_{S^1}^\bullet(Q_1) \right)$$

Theorem – T-Dual Twisted Differential Cohomologies [DMS]
Let $(\mathbb{T}Q_1, H_1)$ and $(\mathbb{T}Q_2, H_2)$ be T-dual twisted standard CAs. Then
$$(d_{H_1}, (-1)^{\dim \mathcal{F}_2} d_{H_2}) U_{R(\Phi)} \subset U_{R(\Phi)}.$$

A Glimpse at GCSs

Definition
A CAR $R: E_1 \dashrightarrow E_2$ is a **generalised complex relation** for generalised complex structures $\mathcal{J}_1, \mathcal{J}_2$ on E_1, E_2 , respectively, if $(\mathcal{J}_1 \times \mathcal{J}_2)(R) = R$, denoted $\mathcal{J}_1 \sim_R \mathcal{J}_2$.

Theorem – T-dual GCSs [DMS]
Let $R(\Phi): \mathbb{T}Q_1 \dashrightarrow \mathbb{T}Q_2$ be a T-duality relation. Given a GCS $\mathcal{J}_1$ with pure spinor ρ , then there exists a pure spinor $\rho_{R(\Phi)}$ defining a GCS $\mathcal{J}_2$ such that $\rho \sim_{U_{R(\Phi)}} \rho_{R(\Phi)}$ and $\mathcal{J}_1 \sim_{R(\Phi)} \mathcal{J}_2$.

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